Joonhyuk Nathanael Kim

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Education

University of California, Irvine — B.S. Computer Engineering — Expected June 2027

Experience

Optum (UnitedHealth Group) — Forward Deployed AI/ML Engineer — **Feb 2026 – Present**

Healthcare technology division processing millions of clinical documents daily

- Built **reusable component system** in TypeScript and React; established design tokens, strict prop typing, and component contracts; system was adopted across 3 internal clinical applications within first sprint cycles
- Leveraged **TypeScript discriminated unions and strict null checks** to enforce data contracts between frontend and backend; reduced classes of runtime errors in clinical data display pipelines
- Built **clinical analytics dashboards** with D3.js and Chart.js; profiled and optimized rendering to maintain sub-200ms load times on datasets with 50K+ daily records

Archv — CEO & Co-Founder — **June 2025 – Present**

AI compliance infrastructure for regulated industries — legal, healthcare, education, government

- Designed and built full **TypeScript + React** frontend from scratch; owned component architecture, state management, routing, and type safety across 12+ product modules spanning the entire application surface
- Implemented **real-time collaborative editing** with WebSockets and conflict resolution for concurrent users; engineered presence indicators, cursor state, and live sync UX with strict TypeScript contracts throughout
- Built **design system** with Tailwind CSS and Storybook; authored typed, accessible components with dark mode and theming support; achieved 95+ Lighthouse scores and full WCAG 2.1 compliance
- Shipped **contract grading product** to production at sub-90s end-to-end latency; secured VP-level interest from UnitedHealth Group and closed 2 active enterprise pilots in legal and healthcare verticals

MedVanta — Software Engineer — **June 2024 – Feb 2025**

Healthcare technology company building clinical operations software for medical practices

- Built **real-time analytics dashboards** using React, D3.js, and WebSockets; achieved sub-100ms rendering on production datasets with accessible, responsive layouts across web and mobile
- Created **component library** of 30+ reusable, typed modules in TypeScript; standardized UI patterns used by all 4 frontend engineers, eliminating duplicated layout and form logic across the codebase
- Owned **responsive layout system** with CSS Grid, Flexbox, and SCSS; implemented cross-browser support and ran direct feedback loops with medical staff to ship iterative improvements
- Developed **native iOS application** in Swift with offline-first architecture connecting patients to 400–600 orthopedic specialists for urgent care, virtual visits, and appointments; designed data sync and notification UX for reliable access across variable connectivity

Technical Leadership

UAV@UCI — President & Technical Lead — Sep 2023 – Sep 2024

- Built **ground station dashboard** in React with real-time telemetry, GPS tracks, and sensor data streaming via WebSockets; engineered smooth rendering of high-frequency data updates in the browser
- Developed **autonomous flight systems** in C++ with PX4 and MAVLink; implemented sensor fusion algorithms combining IMU, GPS, and barometer data for stable navigation in GPS-degraded environments
- Led **25 engineers** across software, electrical, and mechanical sub-teams; established Git workflows, code review standards, and sprint processes across a cross-disciplinary student organization

Projects

VA.gov Modernization MVP — Response to VA RFI for Digital Services Modernization — 2026

- Built 70+ component React/TypeScript frontend replicating core VA.gov workflows in response to an active VA RFI; implemented WCAG 2.1 accessibility standards, responsive layouts, and veteran-facing UX patterns across benefits, healthcare, and records modules
- Authored and submitted formal **RFI response document** outlining system architecture, compliance posture, and modernization approach; proposed a phased migration path with FedRAMP-ready infrastructure on AWS GovCloud